

Lists, S3, & Python Classes

Lecture 05

Dr. Colin Rundel

R Lists

Lists

Lists are R's other vector type (generic). Unlike atomic vectors they are *heterogeneous* - each element can be any R object: atomic vectors, other lists, functions, etc.

```
1 list("A", (1:4)/2, list(1L), sum)
```

R

```
[[1]]  
[1] "A"  
  
[[2]]  
[1] 0.5 1.0 1.5 2.0  
  
[[3]]  
[[3]][[1]]  
[1] 1  
  
[[4]]  
function (..., na.rm = FALSE) .Primitive("sum")
```

```
1 ["A", [0.5, 1.0, 1.5, 2.0], [1], sum]
```

py

```
['A', [0.5, 1.0, 1.5, 2.0], [1], <built-in funct
```

List structure

The printed form of a list is verbose. `str()` gives a compact summary of any R object's structure and is particularly useful for lists.

```
1 str(c(1, 2))
```

```
num [1:2] 1 2
```

```
1 str(1:100)
```

```
int [1:100] 1 2 3 4 5 6 7 8 9 10 ...
```

```
1 str("A")
```

```
chr "A"
```

```
1 str( list(  
2   "A", c(TRUE, FALSE),  
3   (1:4)/2, list(TRUE, 1),  
4   function(x) x^2  
5 ) )
```

List of 5

```
$ : chr "A"
```

```
$ : logi [1:2] TRUE FALSE
```

```
$ : num [1:4] 0.5 1 1.5 2
```

```
$ :List of 2
```

```
..$ : logi TRUE
```

```
..$ : num 1
```

```
$ :function (x)
```

Nested lists

Lists can contain other lists, so they do not have to be flat.

This makes them a natural way of representing tree-like data (e.g. JSON).

```
1 x = list(1, list(2, list(3, 4), 5))  
2 str(x)
```

```
List of 2  
$ : num 1  
$ :List of 3  
..$ : num 2  
..$ :List of 2  
.. ..$ : num 3  
.. ..$ : num 4  
..$ : num 5
```

```
1 x = [1, [2, [3, 4], 5]]  
2 x
```

```
[1, [2, [3, 4], 5]]
```

Named lists

Elements of a list (or atomic vector) can be named. Names help avoid magic numbers when accessing elements (more readable code).

A valid name starts with a letter or `.` (not followed by a digit) and contains only `[A-Za-z0-9._]`.

```
1 x = list(A = 1, B = list(C = 2, D = 3))
2 str(x)
```

```
List of 2
 $ A: num 1
 $ B:List of 2
  ..$ C: num 2
  ..$ D: num 3
```

```
1 names(x)
```

```
[1] "A" "B"
```

Any other name must be surrounded with backticks,

```
1 list("knock knock" = "who's there?")
```

```
$`knock knock`
[1] "who's there?"
```

[vs [[

R has two addition subsetting operators.

- `[[` extracts a *single element*
- `$` is shorthand for named lookup with `[[`

```
1 y = list(a = 1, b = 4, c = 7:9)
```

```
1 y[2]
```

```
$b  
[1] 4
```

```
1 str( y[2] )
```

```
List of 1  
 $ b: num 4
```

```
1 y["b"]
```

```
$b  
[1] 4
```

```
1 str( y["b"] )
```

```
List of 1  
 $ b: num 4
```

```
1 y[[2]]
```

```
[1] 4
```

```
1 str( y[[2]] )
```

```
num 4
```

```
1 y[["b"]]
```

```
[1] 4
```

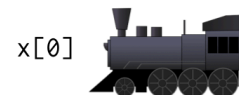
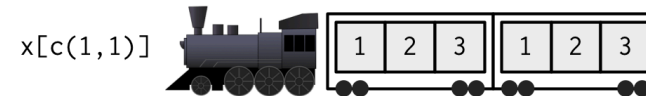
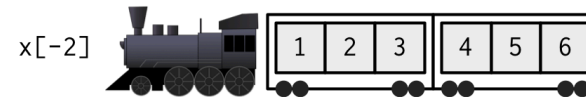
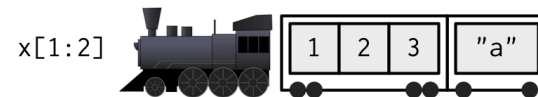
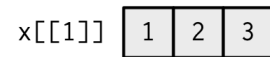
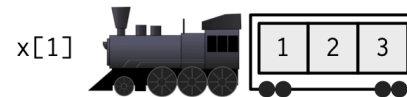
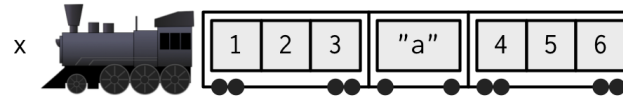
```
1 str( y[["b"]] )
```

```
num 4
```

```
1 y$b
```

```
[1] 4
```

Hadley's analogy



[] details

[] selects by position or by name and only ever returns one element.

```
1 y = list(a = 1, b = 4, c = 7:9)
```

```
1 y[[3]]
```

```
[1] 7 8 9
```

```
1 y[["c"]]
```

```
[1] 7 8 9
```

```
1 y[[4]]
```

```
Error in `y[[4]]`:  
! subscript out of bounds
```

```
1 y[["d"]]
```

```
NULL
```

Vectors of length > 1 are interpreted as *recursive* indexing - `y[[c(3, 2)]]` is `y[[3]][[2]]`.

```
1 y[[c(3, 2)]]
```

```
[1] 8
```

```
1 y[[1:2]]
```

```
Error in `y[[1:2]]`:  
! subscript out of bounds
```

\$ subsetting

`$` is a shorthand for `[[` with a *literal* name, so `y$c` is equivalent to `y[["c"]]`. It only works with lists and objects built using lists (e.g. data frames) and it *partially matches names*.

```
1 y = list(abc = 1, def = 5)
```

R

```
1 y[["abc"]]
```

R

```
[1] 1
```

```
1 y$abc
```

R

```
[1] 1
```

```
1 y$a
```

R

```
[1] 1
```

```
1 y[["a"]]
```

R

```
NULL
```

```
1 x = c(abc = 1, def = 5)
```

R

```
1 x[["abc"]]
```

R

```
[1] 1
```

```
1 x$abc
```

R

```
Error in `x$abc`:  
! $ operator is invalid for atomic vectors
```

A common error is using `$` with a variable that holds a name,

Modifying lists

Assignment with `[[` or `$` replaces an element or adds a new one.

Assigning `NULL` removes an element.

```
1 x = list(a = 1, b = 2)
2 x$c = "new"
3 x[["a"]] = 100
4 str(x)
```

R

List of 3

```
$ a: num 100
$ b: num 2
$ c: chr "new"
```

```
1 x$b = NULL
2 str(x)
```

R

List of 2

```
$ a: num 100
$ c: chr "new"
```

```
1 x = [1, 2]
2 x.append("new")
3 x[0] = 100
4 x
```

py

[100, 2, 'new']

```
1 del x[1]
2 x
```

py

[100, 'new']

Lists and atomic vectors

Combining an atomic vector with a list via `c()` produces a list (the more generic type).

`unlist()` goes the other way, flattening a list into an atomic vector with the usual coercion rules.

```
1 str( c(1, list(4, list(6, 7))) )
```

R

```
List of 3
 $ : num 1
 $ : num 4
 $ :List of 2
  ..$ : num 6
  ..$ : num 7
```

```
1 unlist( list(1:3, list(4:5, 6)) )
```

R

```
[1] 1 2 3 4 5 6
```

```
1 unlist( list(1, list(2, list(3, "Hello"))) )
```

R

```
[1] "1"      "2"      "3"      "Hello"
```

Exercise 1

Represent the following JSON data as a list in R (we will revisit this in Python next week).

```
1 {  
2   "firstName": "John",  
3   "lastName": "Smith",  
4   "age": 25,  
5   "address":  
6   {  
7     "streetAddress": "21 2nd Street",  
8     "city": "New York",  
9     "state": "NY",  
10    "postalCode": 10021  
11  },  
12  "phoneNumber":  
13  [ {  
14    "type": "home",  
15    "number": "212 555-1239"  
16  },  
17  {  
18    "type": "fax",  
19    "number": "646 555-4567"  
20  } ]  
21 }
```

Once you have the list:

- Extract the fax number using `$` and `[[`.
- Add a `"mobile"` phone number.
- Remove the `age` element.

Attributes

Attributes

Attributes are metadata attached to an R object. Some attributes are special (e.g. `names`, `dim`, `dimnames`, `class`, `levels`, etc.) because they modify how the object behaves / is treated.

Attributes are stored as a *named list* attached to the object, accessed via `attributes()` or `attr()`.

```
1 (x = c(L = 1, M = 2, N = 3))
```

```
L M N  
1 2 3
```

```
1 str( attributes(x) )
```

```
List of 1  
 $ names: chr [1:3] "L" "M" "N"
```

```
1 attr(x, "names")
```

```
[1] "L" "M" "N"
```

```
1 attr(x, "other")
```

```
NULL
```

Setting attributes

Most important attributes have helper functions for getting and setting (`names()`, `dim()`, `class()`, `levels()`),

```
1 names(x) = c("Z", "Y", "X")
2 x
```

```
Z Y X
1 2 3
```

```
1 names(x) = 1:3
2 x
```

```
1 2 3
1 2 3
```

```
1 attributes(x)
```

```
$names
[1] "1" "2" "3"
```

```
1 attr(x, "other") = "anything"
2 x
```

```
1 2 3
1 2 3
attr(,"other")
[1] "anything"
```

```
1 str( attributes(x) )
```

```
List of 2
 $ names: chr [1:3] "1" "2" "3"
 $ other: chr "anything"
```

Note that `names(x) = 1:3` silently coerced the integers to character, since the `names` attribute must be a character vector

Factors

Factors are how R represents categorical data - a variable with a discrete set of possible values (called levels).

```
1 (x = factor(c("Sunny", "Cloudy", "Rainy", "Cloudy", "Cloudy")))
```

```
[1] Sunny Cloudy Rainy Cloudy Cloudy  
Levels: Cloudy Rainy Sunny
```

```
1 str(x)
```

```
Factor w/ 3 levels "Cloudy","Rainy",...: 3 1 2 1 1
```

```
1 typeof(x)
```

```
[1] "integer"
```

```
1 mode(x)
```

```
[1] "numeric"
```

```
1 class(x)
```

```
[1] "factor"
```

```
1 levels(x)
```

```
[1] "Cloudy" "Rainy"  "Sunny"
```

Composition

A factor is just an integer vector with two attributes: `levels` and `class`.

```
1 str( attributes(x) )
```

```
List of 2
```

```
$ levels: chr [1:3] "Cloudy" "Rainy" "Sunny"  
$ class : chr "factor"
```

```
1 unclass(x)
```

```
[1] 3 1 2 1 1  
attr("levels")  
[1] "Cloudy" "Rainy" "Sunny"
```

We can build our own from scratch using `attr()`,

```
1 y = c(3L, 1L, 2L, 1L, 1L)  
2 attr(y, "levels") = c("Cloudy", "Rainy", "Sunny")  
3 attr(y, "class") = "factor"  
4 y
```

```
[1] Sunny Cloudy Rainy Cloudy Cloudy  
Levels: Cloudy Rainy Sunny
```

Building objects with `structure()`

Setting attributes one at a time is clunky - `structure()` attaches any number of attributes to an object in a single call and is the standard way to construct objects like this.

```
1 ( y = structure(  
2   c(3L, 1L, 2L, 1L, 1L),  
3   levels = c("Cloudy", "Rainy", "Sunny"),  
4   class = "factor"  
5 ) )
```

```
[1] Sunny Cloudy Rainy Cloudy Cloudy  
Levels: Cloudy Rainy Sunny
```

```
1 class(y)
```

```
[1] "factor"
```

```
1 is.factor(y)
```

```
[1] TRUE
```

```
1 identical(x, y)
```

```
[1] TRUE
```

Factors are integer vectors?

Knowing that factors are stored as integers explains some of their more surprising behaviors,

```
1 x + 1
```

R

Warning in Ops.factor(x, 1): '+' not meaningful for factors

```
[1] NA NA NA NA NA
```

```
1 is.integer(x)
```

R

```
[1] FALSE
```

```
1 is.numeric(x)
```

R

```
[1] FALSE
```

```
1 as.numeric(factor(c("10", "20"))) R
```

```
[1] 1 2
```

```
1 as.integer(x) R
```

```
[1] 3 1 2 1 1
```

```
1 as.character(x) R
```

```
[1] "Sunny" "Cloudy" "Rainy" "Cloudy" "C
```

```
1 as.logical(x) R
```

```
[1] NA NA NA NA NA
```

```
1 as.numeric( R
2   as.character(factor(c("10", "20")))
3 )
```

```
[1] 10 20
```

S3 Object System

class

The `class` attribute adds a layer on top of R's type hierarchy - previously we saw `typeof()` and `mode()`.

value	<code>typeof()</code>	<code>mode()</code>	<code>class()</code>
<code>TRUE</code>	logical	logical	logical
<code>1</code>	double	numeric	numeric
<code>1L</code>	integer	numeric	integer
<code>"A"</code>	character	character	character
<code>NULL</code>	NULL	NULL	NULL
<code>list(1, "A")</code>	list	list	list
<code>factor("A")</code>	integer	numeric	factor
<code>matrix(1:4, 2)</code>	integer	numeric	matrix, array
<code>function(x) x^2</code>	closure	function	function
<code>sum</code>	builtin	function	function

S3 class specialization

```
1 x = c("A", "B", "A", "C")
```

R

```
1 print( x )
```

R

```
[1] "A" "B" "A" "C"
```

```
1 print( factor(x) )
```

R

```
[1] A B A C  
Levels: A B C
```

```
1 print( unclass( factor(x) ) )
```

R

```
[1] 1 2 1 3  
attr("levels")  
[1] "A" "B" "C"
```

```
1 print.default( factor(x) )
```

R

```
[1] 1 2 1 3
```

What's up with `print`?

```
1 print
```

R

```
function (x, ...)  
UseMethod("print")  
<bytecode: 0x75fffdccb8>  
<environment: namespace:base>
```

`print` does no printing itself - `UseMethod()` looks at the class of `x` and calls the matching *method*. For a factor that is `print.factor()`, for everything without a more specific method `print.default()` is used.

1 print.default

R

```
function (x, digits = NULL, quote = TRUE, na.pri
  right = FALSE, max = NULL, width = NULL, use
  ...)
{
  args <- pairlist(digits = digits, quote = qu
    print.gap = print.gap, right = right, ma
    useSource = useSource, ...)
  missings <- c(missing(digits), missing(quote
    missing(print.gap), missing(right), miss
    missing(useSource))
  .Internal(print.default(x, args, missings))
}
<bytecode: 0x75fcd11968>
<environment: namespace:base>
```

1 print.factor

R

```
function (x, quote = FALSE, max.levels = NULL, w
  ...)
{
  ord <- is.ordered(x)
  if (length(x) == 0L)
    cat(if (ord)
      "ordered"
      else "factor", "()\n", sep = "")
  else {
    xx <- character(length(x))
    xx[] <- as.character(x)
    keepAttrs <- setdiff(names(attributes(x)
      "class"))
    attributes(xx)[keepAttrs] <- attributes(
      print(xx, quote = quote, ...))
  }
  maxl <- max.levels %||% TRUE
  if (maxl) {
    n <- length(lev <- encodeString(levels(x)
      "\'", "'"))
    colsep <- if (ord)
      " < "
    else " "
```

Generics are everywhere

Many of the base R functions you use regularly are S3 generics whose only job is to dispatch on class,

1 mean

R

```
function (x, ...)  
  UseMethod("mean")  
<bytecode: 0x75fe471738>  
<environment: namespace:base>
```

1 plot

R

```
function (x, y, ...)  
  UseMethod("plot")  
<bytecode: 0x75f694dc08>  
<environment: namespace:base>
```

1 summary

R

```
function (object, ...)  
  UseMethod("summary")  
<bytecode: 0x75f80fca88>  
<environment: namespace:base>
```

1 t.test

R

```
function (x, ...)  
  UseMethod("t.test")  
<bytecode: 0x75ff1b1150>  
<environment: namespace:stats>
```

Other generics, such as `sum`, dispatch without an explicit call to `UseMethod()`:

1 sum

R

```
function (... , na.rm = FALSE) .Primitive("sum")
```

`sum` is a primitive implemented in C. It still performs S3 dispatch internally, rather than calling `UseMethod()` in its visible

What is S3?

S3 is R's first and simplest OO system. S3 is R's first and simplest OO system. S3 is informal and ad hoc, but there is a certain elegance in its minimalism: you can't take away any part of it and still have a useful OO system.

- Hadley Wickham, Advanced R

Dispatch

S3 dispatch is very simple - a generic calls `UseMethod("func")`, which searches for a function named `<func>.<class>`, trying each class of the first argument in order and falling back to `<func>.default` if none is found.

`methods()` lists the methods available for a generic, or all the methods available for a class,

```
1 methods("summary")
```

```
[1] summary,ANY-method          summary,diagonalMatrix-method
[3] summary,sparseMatrix-method summary.aov
[5] summary.aovlist*            summary.aspell*
[7] summary.check_packages_in_dir* summary.connection
[9] summary.data.frame          summary.Date
[11] summary.default             summary.difftime
[13] summary.ecdf*               summary.factor
[15] summary.free1way*           summary.glm
[17] summary.infl*               summary.lm
[19] summary.loess*              summary.manova
[21] summary.matrix              summary.mlm*
[23] summary.nls*                summary.packageStatus*
[25] summary.pandas.core.frame.DataFrame* summary.pandas.core.series.Series*
[27] summary.pandas.DataFrame*   summary.pandas.Series*
[29] summary.POSIXct             summary.POSIXlt
[31] summary.ppr*                summary.prcomp*
[33] summary.princomp*           summary.proc_time
```

[35]	summary.python.builtin.object*	summary.rlang_error*
[37]	summary.rlang_message*	summary.rlang_trace*
[39]	summary.rlang_warning*	summary.rlang:::list_of_conditions*
[41]	summary.shingle*	summary.srcfile
[43]	summary.srcfile	summary.stepfun
[45]	summary.table	summary.table

```
1 methods(class = "factor")
```

[1]	[	[[	[[<-	[<-	all.equal
[6]	Arith	as.character	as.data.frame	as.Date	as.list
[11]	as.logical	as.POSIXlt	as.vector	c	cbind2
[16]	coerce	Compare	droplevels	format	free1way
[21]	initialize	is.na<-	kronecker	length<-	levels<-
[26]	Logic	Math	Ops	plot	print
[31]	rbind2	relevel	relist	rep	show
[36]	slotsFromS3	summary	Summary	xtfrm	

see '?methods' for accessing help and source code

Adding methods

Because dispatch is by *name*, adding a method for a new class just requires defining a new function with the proper name - no registration is needed.

```
1 ( x = structure(  
2   c(1, 2, 3),  
3   class = "class_A") )
```

R

```
[1] 1 2 3  
attr(,"class")  
[1] "class_A"
```

```
1 ( y = structure(  
2   c(6, 5, 4),  
3   class = "class_B") )
```

R

```
[1] 6 5 4  
attr(,"class")  
[1] "class_B"
```

```
1 print.class_A = function(x, ...) {  
2   cat("(Class A) ")  
3   print.default(unclass(x))  
4 }
```

R

```
1 print.class_B = function(x, ...) {  
2   cat("(Class B) ")  
3   print.default(unclass(x))  
4 }
```

R

```
1 print(x)
```

R

```
(Class A) [1] 1 2 3
```

```
1 print(y)
```

R

```
(Class B) [1] 6 5 4
```

```
1 class(x) = "class_B"  
2 print(x)
```

R

```
(Class B) [1] 1 2 3
```

```
1 class(y) = "class_A"  
2 print(y)
```

R

```
(Class A) [1] 6 5 4
```

Writing a good `print` method

In general, methods should match the generic's signature (`x, ...`). For `print` you should then manage output via `cat()` or `print()`, and return the input *invisibly* so that `print(x)` can be used in a pipeline.

```
1 pct = structure(c(0.12, 0.5, 1), class = "percent")
2
3 print.percent = function(x, ...) {
4   print(paste0(unclass(x) * 100, "%"), quote = FALSE)
5   invisible(x)
6 }
```

```
1 pct
```

```
[1] 12% 50% 100%
```

```
1 z = print(pct)
```

```
[1] 12% 50% 100%
```

```
1 unclass(z)
```

```
[1] 0.12 0.50 1.00
```

This goes for any new method - match the signature and conform to the common expectations of the other method

Defining a new generic

A generic is just a function that calls `UseMethod()`. By convention the first argument is the object dispatched on and `...` is included so that methods can add arguments.

```
1 shuffle = function(x, ...) {  
2   UseMethod("shuffle")  
3 }
```

R

```
1 shuffle.default = function(x, ...) {  
2   stop("Class ", class(x), " is not supported by shuffle.", call. = FALSE)  
3 }
```

R

```
1 shuffle.factor = function(x, ...) {  
2   factor( sample(as.character(x)), levels = sample(levels(x)) )  
3 }
```

R

```
1 shuffle.integer = function(x, ...) {  
2   sample(x)  
3 }
```

R

Shuffle results

```
1 shuffle( 1:10 )
```

R

```
[1] 6 8 9 3 4 1 2 5 7 10
```

```
1 shuffle( factor(c("A", "B", "C", "A")) )
```

R

```
[1] C A A B
```

```
Levels: A C B
```

```
1 shuffle( c(1, 2, 3, 4, 5) )
```

R

Error:

! Class numeric is not supported by shuffle.

```
1 shuffle( letters[1:5] )
```

R

Error:

! Class character is not supported by shuffle.

```
1 methods("shuffle")
```

R

```
[1] shuffle.default shuffle.factor shuffle.integer  
see '?methods' for accessing help and source code
```

Implicit classes

Objects without a `class` attribute dispatch on an *implicit* class vector that is more detailed than what `class()` reports.

```
1 report = function(x) {  
2   UseMethod("report")  
3 }  
4 report.default = function(x) paste0("Class ", class(x), " does not have a method defined.")  
5 report.integer = function(x) "I'm an integer!"  
6 report.double = function(x) "I'm a double!"  
7 report.numeric = function(x) "I'm a numeric!"
```

```
1 report(1)
```

```
[1] "I'm a double!"
```

```
1 report(1L)
```

```
[1] "I'm an integer!"
```

```
1 report("1")
```

```
[1] "Class character does not have a method defined."
```

```
1 rm(report.integer, report.double)  
2 report(1)
```

```
[1] "I'm a numeric!"
```

```
1 report(1L)
```

```
[1] "I'm a numeric!"
```

```
1 rm(report.numeric)  
2 report(1)
```

```
[1] "Class numeric does not have a method defined."
```

```
1 report(1L)
```

```
[1] "Class integer does not have a method defined."
```

Why?

From [UseMethod](#)'s R documentation:

If the object does not have a class attribute, it has an implicit class. Matrices and arrays have class “matrix” or “array” followed by the class of the underlying vector. Most vectors have class the result of `mode(x)`, except that integer vectors have class `c("integer", "numeric")` and real vectors have class `c("double", "numeric")`.

The implicit class vector can be inspected with `.class2()`,

```
1 .class2(1)
```

```
[1] "double" "numeric"
```

```
1 .class2(1L)
```

```
[1] "integer" "numeric"
```

```
1 .class2(matrix(1:4, 2))
```

```
[1] "matrix" "array" "integer" "numeric"
```

`report(1)` tries `report.double`, then `report.numeric`, then `report.default` - which is why a method for `double` is found even though `class(1)` says `"numeric"`.